

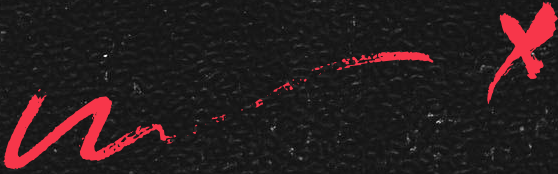


# Types of Evidence

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# Witness Testimony





# Who can be a witness?

"A person who saw or heard the crime take place or may have important information about the crime or defendant"

US Department of Justice

- Can be called by prosecution or defense
- What the witness says is called **testimony**
- The witness shares what they know about the situation





## Lay witness

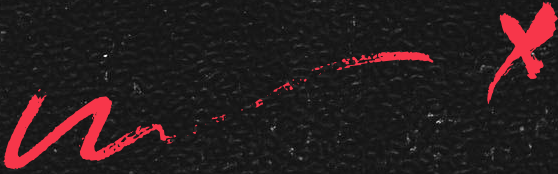
Witnessed the events,  
describes what they saw

## Character witness

Knows someone involved in the case,  
may speak to the character of the  
victim or defendant

## Expert witness

An expert in their field,  
testifies on their specialty  
only



# Physical Evidence





# What is Physical Evidence?

It's anything that can be seen, smelled, touched, or collected from a crime scene including objects that can be identified through the senses and are used in court to prove facts



Used to identify suspects and victims while also determining how the crime was committed.

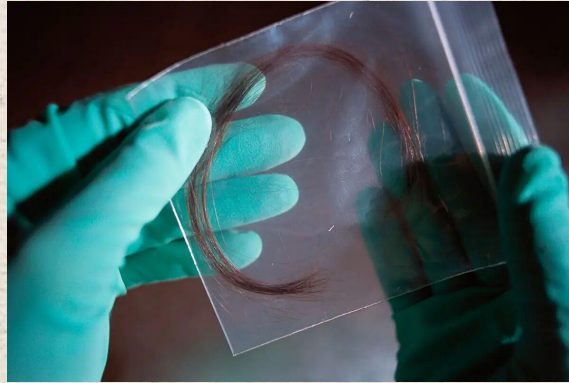
**3 Main types:**  
biological, impressions,  
trace evidence.



# First Type of Evidence

**Biological:** Has to do with any biological matter

- Blood
- Saliva
- Sweat
- Hair
- Semen
- Skin Cells

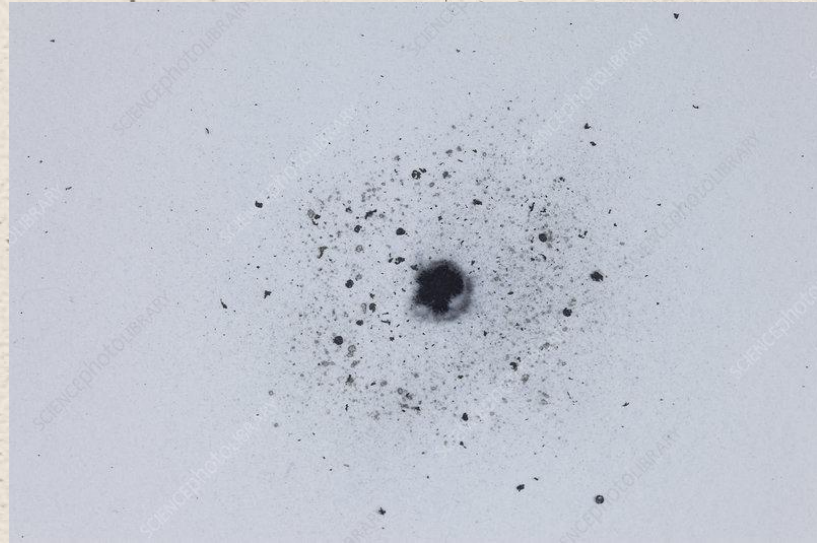




# Second Type of Evidence

**Trace:** materials transferred between people, objects, or locations during a crime

- Fiber
- Soil
- Gunpowder Residue
- Pollen
- Glass Fragments
- Paint chips



# Third Type of Evidence

**Impressions:** any mark left by an object after it has made contact with another surface

- Fingerprints
- Footprints
- Tire tracks
- Tool marks
- Bite marks





# How does it work:

## In crime scene

- Investigators locate potential evidence with gloves, lights, photography, and field notes.
- Items not to be moved from their original position.

## In forensics

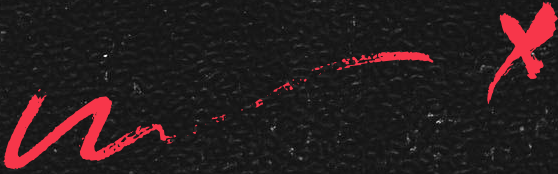
- Evidence is collected with swabs, tweezers and containers.
- It is sealed in bags/envelopes and labeled with date, time, location, and the collector's name.

## In custody

- Each piece gets a unique ID
- A record is kept of every person who handles the evidence.
- Ensures proof that nothing was altered or tampered with.

## In court

- Forensic experts present lab results.
- Evidence is used to link a suspect, reconstruct what happened, or prove someone innocent.



# Digital Evidence





# Understanding the Basics: Evidence vs. Crime

## Digital Forensics

- The process of finding, identifying, extracting, analyzing, and recording digital evidence.
- The science of finding evidence from digital media.
- The goal is to identify, preserve, and analyze the evidence in a way that is admissible in court.

## Digital Evidence

- Digital evidence is information that helps establish the fact that the crime has been committed or establishes links between crime, victim, and culprit.
- Digital evidence is abundant and powerful, but the ease of change makes it fragile and vulnerable to claims of errors, alteration, and fabrication.

## Cybercrime

- Criminal activity that targets or uses computers, networks, or digital systems as the primary tool of the offense.
- Cybercrimes may involve hacking, identity theft, malware, online fraud, or cyberstalking.
- The defining feature is that the crime could not exist without digital technology.

**Investigator  
Responsible  
Evidence**

# Who handles digital evidence?

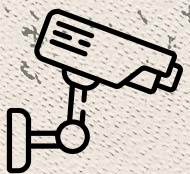


Collect and  
secure devices  
on scene.

Image, extract, and  
analyze digital  
evidence in a lab.

Request specific  
information pertaining  
to digital evidence to  
help build their case.





# What is Digital Evidence?



Metadata

Video Surveillance

Smartwatches

Social Media Activity



Internet Search History

Timestamps

Computer Files

Photos

Call Logs

Smart Home Devices

Spreadsheets

App Data

Google Drive

Deleted Files

Geolocation

GPS

Emails

Texts

Documents

Device Identifiers

Audio Recording

Dropbox

Fitness Trackers

iCloud

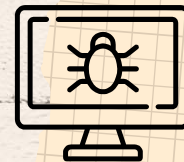
Game consoles

Financial & Transactional Records

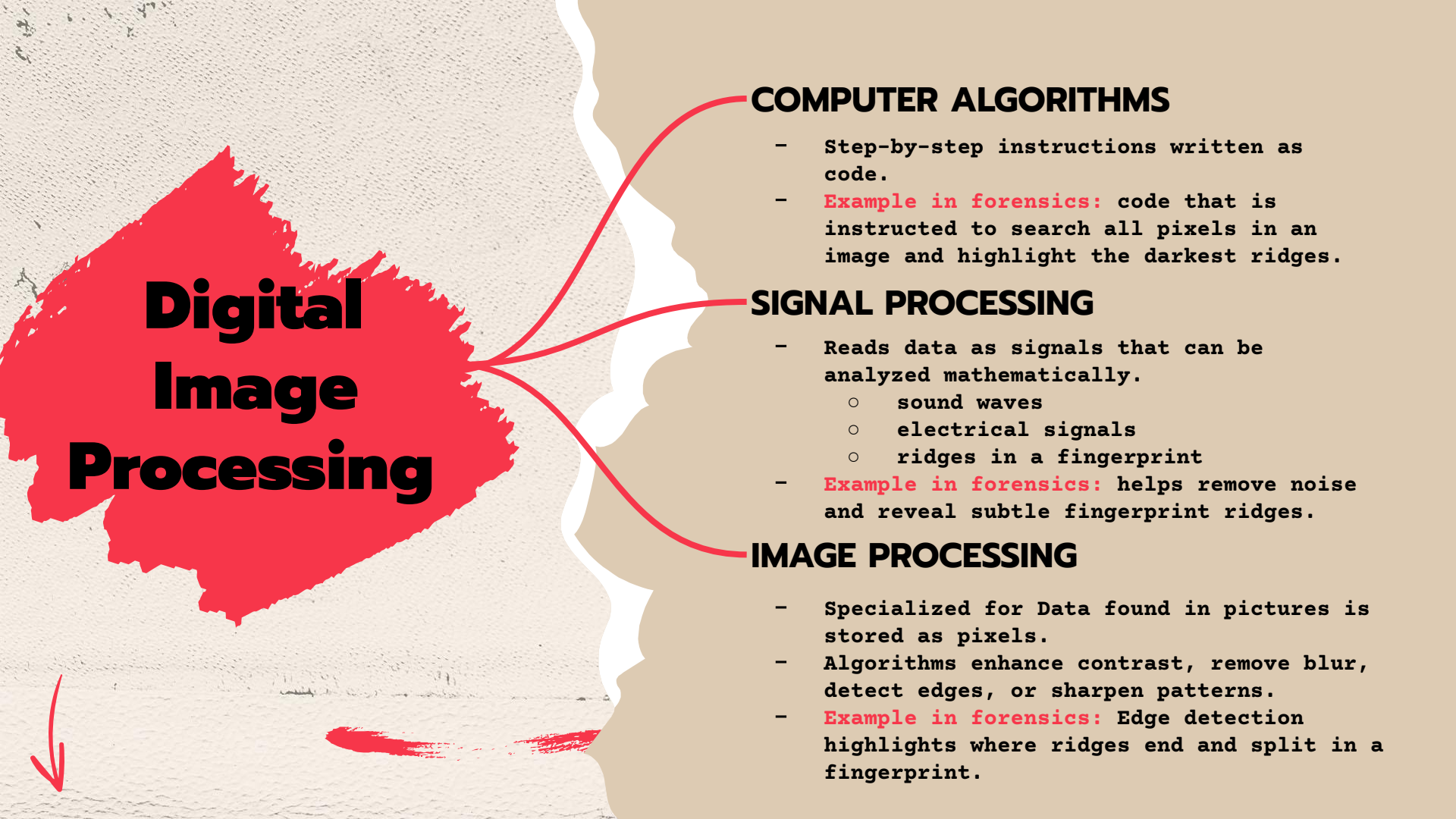
Biometric Systems

VoIP

Network Logs



# Digital Image Processing



## COMPUTER ALGORITHMS

- Step-by-step instructions written as code.
- **Example in forensics:** code that is instructed to search all pixels in an image and highlight the darkest ridges.

## SIGNAL PROCESSING

- Reads data as signals that can be analyzed mathematically.
  - o sound waves
  - o electrical signals
  - o ridges in a fingerprint
- **Example in forensics:** helps remove noise and reveal subtle fingerprint ridges.

## IMAGE PROCESSING

- Specialized for Data found in pictures is stored as pixels.
- Algorithms enhance contrast, remove blur, detect edges, or sharpen patterns.
- **Example in forensics:** Edge detection highlights where ridges end and split in a fingerprint.



# How Digital Evidence Is Used

DETERMINING  
MOTIVE

MATCH LOCATION  
DATA TO THE  
CRIME SCENE

GATHER  
FINANCIAL  
ACTIVITY

PROVING  
PREMEDITATION

BUILDING A  
TIMELINE

CORROBORATE OR  
CONTRADICT WITNESS  
TESTIMONY

LINKING SUSPECTS,  
VICTIMS, AND  
ACCOMPLICES

CONFIRMING  
OR  
DISPROVING  
AN ALIBI



# Common Challenges

## Encryption

Encryption is utilized by criminals to conceal evidence. Analysts try to decrypt the materials but because attackers use a wide variety of encryption techniques, it takes a lot of time, and occasionally, it is impossible to decrypt.

## Data Silos

When evidence is stored across multiple disconnected systems, it becomes difficult to access and analyze everything.

## Volume & Diversity

The volume and diversity of digital evidence found in each case is increasing. If departments do not have enough trained professionals to process the large volume, the result is a backlog no matter what tools are used.

## Evolving Technology

As technologies like AI, blockchain, 5G, and cloud computing evolve, they offer new opportunities but also introduce significant challenges. Analysts must stay up to date in order to effectively collect, analyze, and present digital evidence.

## Chain of Custody

Maintaining an unbroken chain of custody is vital because any break or inconsistency can lead to challenges regarding the authenticity. Digital evidence can be easily manipulated or corrupted making this even more critical. The chain of custody ensures that digital evidence is admissible in court and holds up against scrutiny.



# Conclusion

Evidence comes in a variety of shapes and forms. It can take investigators hours, days, months, or even years just to find a single piece of important evidence. So don't believe everything you see on TV because that leads to misinformation, making it harder for first responders and investigators to do their jobs.

# Class Activity

Case File -

April 15, 2013 | Boston Marathon Bombing

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Two homemade bombs exploded near the finish line of the Boston Marathon, killing 3 people and injuring hundreds. Investigators collected different types of evidence to identify and prosecute the suspects.

Please refer to the instructions contained inside of the bag.



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